Roll No. | | | | | | | |

END SEMESTER EXAMINATION JUNE- 2025

Name of The Program: ***B.Tech (Petroleum Engineering)***
Name of The Course: **Reservoir Engineering**

Semester: IV
Course Code: **TPE-402**

Time: 3 Hours

Maximum Marks: 100

Note:
(i) All Questions are compulsory
(ii) Answer any two questions among a, b & c in each main question.
(iii) Total marks each main question is twenty.
(iv) Each Question carries 10 Marks

Q1. **(20 Marks)**
(CO2,3)

a) i) Describe the concept of reservoirs and reservoir engineering. distinguish between conventional and unconventional reservoir
ii) Write a brief note
1. Cricondenbar
2. Dew Point Curve
3. Quality Lines
4. Phase Envelope

b) i) If a well is having limited perforated interval and partially penetrated pay zone then explain the types of flow geometry.
ii) Describe different types of flow regimes.

c) Explain Any 8 Types of fluid flow.

Q2. **(20 Marks)**
(CO1,2)

a) Differentiate between Undersaturated, saturated oil reservoirs, wet and dry gas reservoirs.

b) An oil reservoir exists at its bubble-point pressure of 3100 psia and temperature of 170°F. The oil has an API gravity of 45° and gas-oil ratio of 650 scf/STB. The specific gravity of the solution gas is 0.67.
The following additional data are also available:
Reservoir area = 680 acres
Average thickness = 15 ft
Connate water saturation = 0.28
Effective porosity = 18%
Calculate the initial oil in place in STB.

c) i) Calculate the pressure difference, i.e., capillary pressure, and capillary rise in an oil-water system from the following data:
Contact Angle = 30°
Water density = 1.0 gm/cm^3
Oil density= 0.75 gm/cm^3
r = 10^{-4} cm
Oil water surface tension = 25 dynes/cm

ii) A brine is used to measure the absolute permeability of a core plug. The rock sample is 4 cm long and 3 cm^2 in cross section. The brine has a viscosity of 1.0 cp and is flowing a constant rate of 0.5 cm^3/sec under a 2.0 atm pressure differential. Calculate the absolute permeability.

Q3. (20 Marks) (CO2,3)

a) Distinguish between Gas Cap Drive, Water Drive and Gravity Drive.

b) Describe Reservoir rock, its permeability and porosity, and distinguish between primary and secondary porosity

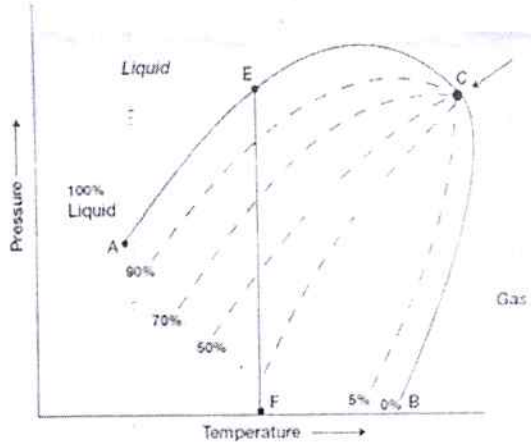
c) Write a Brief Note on: -
   i) Critical oil Saturation
   ii) Critical Gas Saturation
   iii) Residual Oil Saturation
   iv) Movable Oil Saturation
   v) Critical Water Saturation

Q4. (20 Marks) (CO4)

a) Explain all the parameters affecting porosity and permeability.
b) With the help of a diagram explain the concept of wettability and capillary pressure.
c) Explain Darcy's law and describe radial, liner flows with diagram.

Q5 (20 Marks) (CO5)

a) In the given figure below explain Points (C, E, F), line (AC, BC, BCA) and quality Lines



b) Describe structural, stratigraphic & Combinations traps.
c) Explain the concept of Gas and water conning